CHILDHOOD TRAUMA LEGACY: EMOTIONAL MEMORY AND NEURAL RESPONSES TO 9/11-RELATED STIMULI

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Introduction

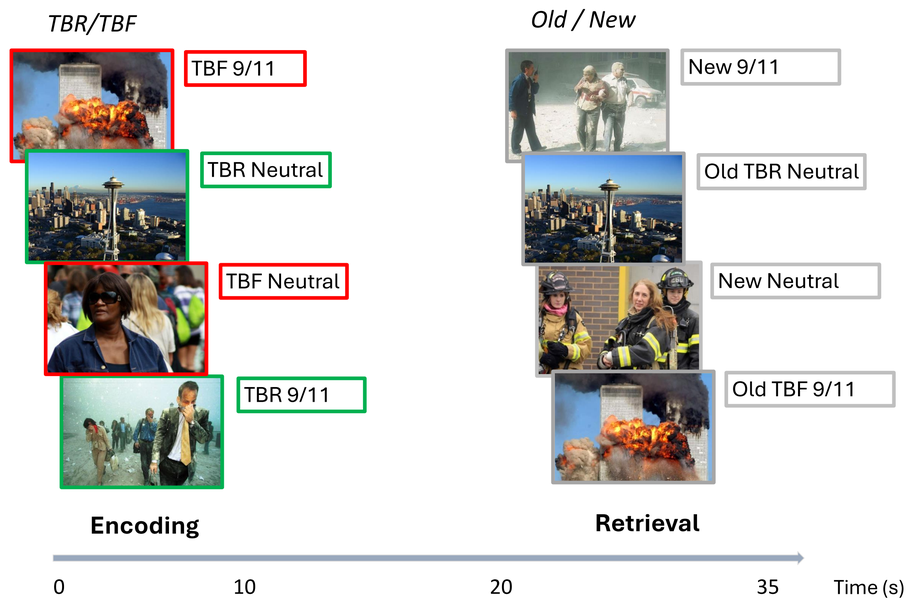
- Children and adolescents exposed to the 9/11 WTC attack continue to face neuropsychological and emotional challenges in adulthood.
- 15 years after 9/11, exposed children were more likely to have mental health disorders compared to non-exposed peers.
- Mental health disorders are associated with deficits in neuronal networks involved in emotion regulation, cognitive control and memory.
- Neurobiological circuits underlying regulation, control and memory undergo significant development during childhood and adolescence, making these periods especially vulnerable to traumatic experiences.

Aim

Identify differences between exposure groups on behavioral and temporal brain signals elicited during cognitive control of memory processing.

Directed Forgetting Task Paradigm

Participants were instructed to remember the scene with **green** frame (To be Remembered, TBR) and not to remember those with **red** frame (To be Forgotten, TBF) During retrieval, each stimulus was presented for 1 s and trial duration was 2.5 s.



Stimuli used in Encoding and Retrieval Phases.

Stimulus Type	Encoding Instructions		Retrieval Phase			
	TBR	TBF	TBR Old	TBF Old	New	Total
9/11	40	40	40	40	80	160
Neutral	40	40	40	40	80	160

Sample

57 adults, mean age 33 years (range 26-39), ages 5-18 years old on 9/11/2001; 32 F
3 groups did not differ in demographic characteristics.

- exposed with mental health disorders – N=21] **Mental Health (MH) group**
- exposed without mental health disorders – N= 23] **Healthy Control (HC) group**
- not exposed without mental health disorders – N=13

EEG Recordings

Continuous 64-channel EEG data recorded using the extended 10-20 system, actiChamp EEG system (Brain Vision) (DC; 250 Hz upper cutoff; 500 Hz digitization rate). The data includes left and right mastoids, referred to Cz. Vertical and horizontal EOG were recorded and used to remove artifacts.

Analyses

Behavioral - memory performance (i.e., hit – false alarm rates), bias, and mean RTs were calculated for each stimulus type for each participant.

EEG – ERP voltage averages at 9 electrode sites were calculated for 11 time-intervals of 40 ms each from 200 ms to 660 ms, for correct responses for each stimulus type and for each participant during the retrieval phase.

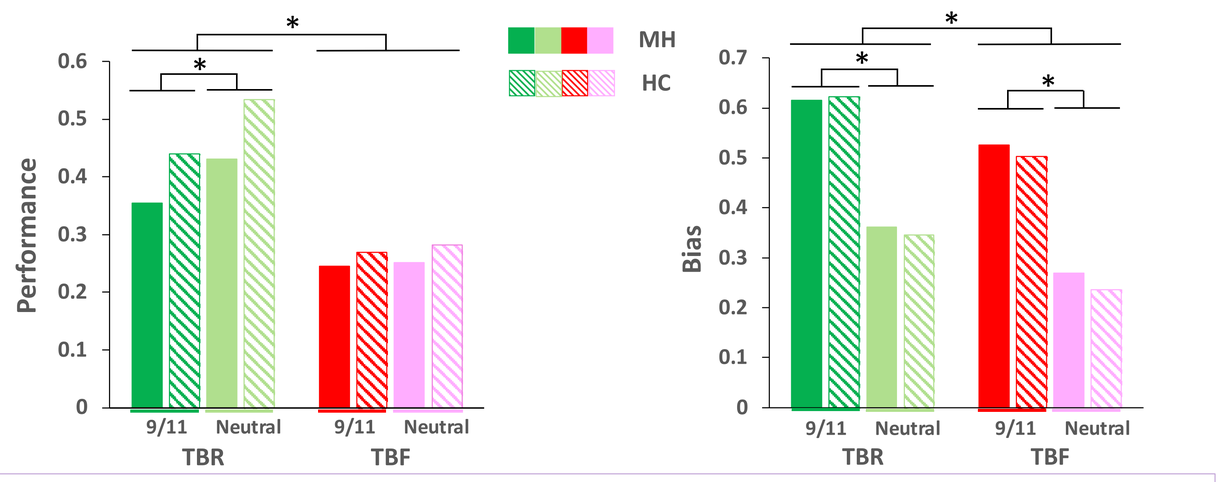
Statistics - Repeated mixed design ANOVA was used to compare between groups and conditions for behavioral and EEG data

Acknowledgments

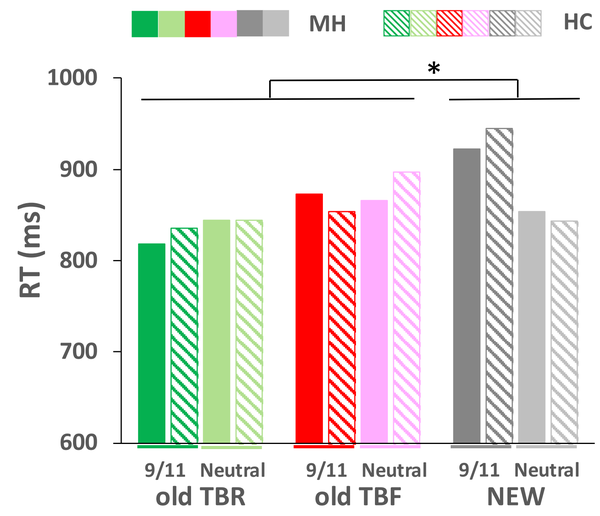
Thanks to all the participants for their time, to the GPEG team for their support with recruitment, and to Charlie Brown III for technical support. Thanks for support from the NIOSH/CDC grant U01OH011694. **Contact: yc60@columbia.cumc.edu**

Results

Memory Performance during Retrieval

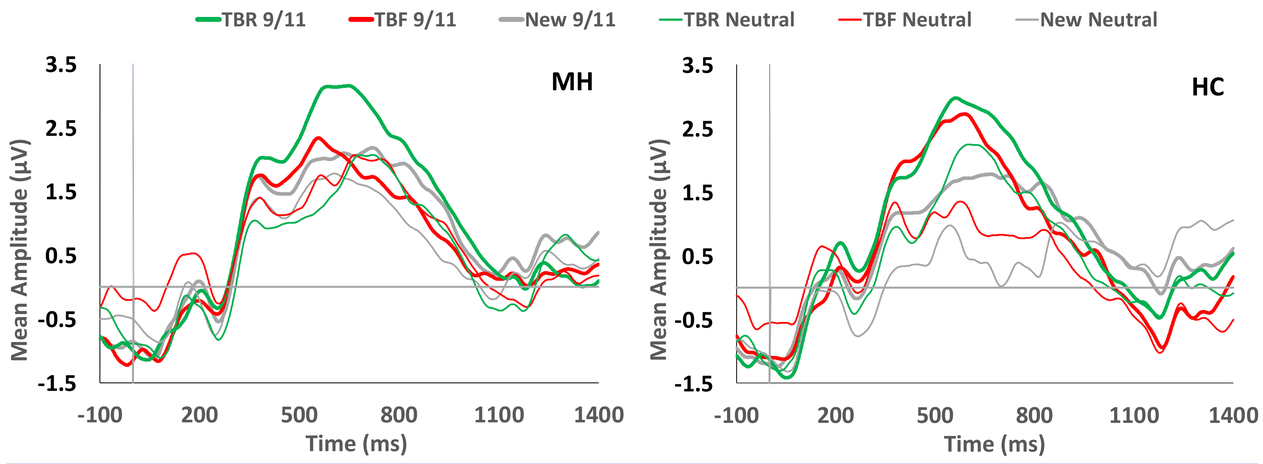


- Performance differed between TBR and TBF ($F_{1,55}$=60.15, p<0.0001).
- Performance was higher for neutral than for 9/11-related stimuli ($F_{1,55}$=13.10, p<0.001). There was a significant interaction between stimulus type and stimulus instruction ($F_{1,55}$=12.29, p<0.001).
- Performance was similar across groups ($F_{1,55}$=2.98, p<0.09).
- There was a significant performance bias between the two stimulus types ($F_{1,55}$=54.59, p<0.001), and across stimulus instruction ($F_{1,55}$=95.18, p<0.001) but no significant differences in bias across groups.



- RT was significantly different for old vs. new stimuli (old/new effect), with old TBR being the fastest ($F_{2,110}$=12.82, p<0.0001)
- 3 ways Interaction of Group x Stimuli x Instruction ($F_{2,110}$=5.60, p<0.004) revealed that TBR 9/11 stimuli was faster but New 9/11 was slower with TBF 9/11 also slower for the MH group.

ERP - Old / New effects (amplitude for old > new) at Pz electrode site



- Main effects old/new ERP effect were significant (ps<0.03), with larger amplitudes for old TBR, followed by that of old TBF images.
- Main effects of Stim (ps<0.01) were significant indicting larger amplitudes for 9/11 than neutral stimuli.
- Significant Stim x Group ($F_{1,52}$=25.77, p<0.02) and Stim x Cond ($F_{2,104}$=3.14, p<0.05) interactions indicate that in the MH group an old/new effect was seen only for TBR 9/11-related images, and weaker and less differentiating effect for the TBR Neutral and TBF stimuli. In contrast, the HC group showed a clear old/new ERP effect for all stimuli types.

Conclusions

- Although the task was difficult, performance was better for TBR than for TBF stimuli.
- Neutral stimuli were remembered better possibly because they were more distinct than 9/11-related stimuli.
- The groups did not differ in Performance, but their ERP responses were clearly different.
- HC group showed larger amplitudes for TBR stimuli, and a similar old/new effect for both 9/11 and neutral stimuli. There were overall larger amplitudes for the emotional 9/11 stimuli than for the neutral stimuli.
- MH group showed a noticeable larger amplitude for the TBR 9/11 stimuli only. This suggest that the MH group was fully engaged in retrieval of TBR 9/11 stimuli, but their responses on other stimuli was impaired, potentially reflecting their lower (but not significant) behavioral performance which may have been driven by more correct guesses.